

# SAHAKAR SETU — Grievance Module (Detailed)

How a spoken complaint becomes a formal, addressed, trackable application — in Hindi, without the member writing a single word.

## 1. The Problem This Solves

A rural PACS member with a stuck PMFBY claim or a loan dispute cannot draft a formal application: they don't know the format, the correct addressee, or the legal language. Sahakar Setu interviews them conversationally (on any channel), extracts the facts, and generates a formal grievance application addressed to the **correct authority** — with a tracking ID they can check from a feature phone.

## 2. Eight Grievance Categories → Correct Addressee

| Category         | Addressee (Hindi template)                 |
|------------------|--------------------------------------------|
| society_pacs     | PACS Secretary / PACS Prabandhak           |
| cooperative_bank | Bank Nodal Grievance Officer               |
| insurance_pmfby  | Insurance Grievance Cell / Bima Lokpal     |
| registrar_level  | Registrar of Cooperative Societies         |
| election         | Registrar / Cooperative Election Authority |
| loan             | Branch Manager, Sahakari Bank              |
| subsidy          | District Cooperative Officer               |
| land             | Tehsildar / Sub-Divisional Magistrate      |

## 3. Fact Memory Feeds the Template

Conversation turns are mined for structured facts ( `extractFacts` in `src/modules/case-memory/` ): **name, father's name, village, district, state, PACS name, membership number, issue description, application date, receipt number, amount, phone**. Facts accumulate in the case ( `case_facts` table) across calls and channels, so the grievance draft fills itself without re-asking. (The code uses camelCase `factKey/factValue` ; Supabase stores snake\_case `fact_key/fact_value` — mapped in `getCaseFacts` .)

## 4. Draft Generation (Handlebars, Hindi-first)

`generateDraft` ( `src/modules/grievance/index.ts` ) compiles `templates/grievance.hi.hbs` with the facts. A generated Hindi draft looks like:

सेवा में,  
श्रीमान सचिव, प्राथमिक कृषि साख समिति (PACS Secretary)

**विषय:** PMFBY शिकायत: गेहूं की फसल का नुकसान

महोदय,

मैं, राम लाल, ग्राम सरायकेला का सदस्य (सदस्यता सं. 4521) हूँ। मेरी गेहूं की फसल ओलावृष्टि में नष्ट हो गई है। मैंने दिनांक 12-03-2026 को आवेदन किया था, रसीद सं. 88412। अब तक दावा लंबित है।

कृपया उपरोक्त मामले का निस्तारण 15 दिनों के भीतर करने की कृपा करें।

सादर,  
राम लाल

दिनांक: 28-09-2026

**ट्रैकिंग आईडी: SS-2026-000006**

## 5. Tracking IDs That Survive

- Format: `SS-2026-XXXXXX` (e.g. the live-verified `SS-2026-000006` ).
- Collision-safe: ID generation retries up to 3× if the ID already exists in `grievance_drafts` .
- Stored in `grievance_drafts` (case\_id, tracking\_id, addressee, subject, body, language) + a `grievance_tracking` status row ( `drafted` ).
- Track on any channel: `GET /api/v1/grievance/track/:trackingId` returns the draft + full status history (newest first).

## 6. Time-Based Escalation Ladder

`getEscalationPath` ( `src/modules/escalation/` ) computes the member's current step from the case age, with a ready draft template at every level:

| Step | Authority                          | Triggers                                | Draft template                     |
|------|------------------------------------|-----------------------------------------|------------------------------------|
| 1    | Society / PACS Secretary           | Immediately (status <code>open</code> ) | <code>society_grievance</code>     |
| 2    | Registrar of Cooperative Societies | > 15 days without resolution            | <code>registrar_grievance</code>   |
| 3    | Insurance Ombudsman                | PMFBY cases > 30 days                   | <code>ombudsman_complaint</code>   |
| 4    | Legal Aid (NALSA)                  | Escalated cases > 60 days               | <code>legal_aid_application</code> |

## 7. Honest Case Strength (Verdict Engine)

- Each case gets a **0–100 strength score** with the missing evidence listed — no false hope.
- Cases scoring **60+** are offered verified-advocate connect ( `/api/v1/lawyer/connect` ).

- The score comes from facts present vs. facts required for that grievance category (documents, dates, receipts).

## 8. Kiosk Printout

On the kiosk (Capacitor Android app), the draft and tracking receipt print on a 58/80 mm ESC/POS thermal printer via USB OTG — the member walks away with a signed-form-ready paper application. On the web/PWA the same draft is copyable and downloadable.

**Live-verified:** grievance draft + tracking flow tested end-to-end on the deployed server — Hindi template fills facts correctly, tracking ID retrievable.